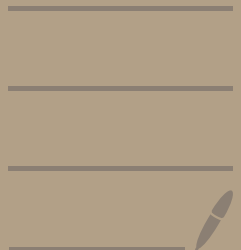


Arrays



Arrays

Definition of Array

An array is a "Data structure" containing a "number of data values" (all of which are of "Same type")

What is a Data Structure

Data structure is a format for organizing and storing data.

→ Also, each data structure is designed to organize data to suite a specific purpose.

Example: Array is a data structure which you can visualize as follows.



Imagine an array as a large chunk of memory divided into smaller blocks of memory and each block is capable of storing a data value of some type.

A	5	6	10	-1	7	8	101	97	88	67
---	---	---	----	----	---	---	-----	----	----	----

This array consists of 10 data values

A

5	6	10	13	56	76	1	2	4
---	---	----	----	----	----	---	---	---

 ✓

B

'a'	'b'	'c'	'd'	'e'	'f'
-----	-----	-----	-----	-----	-----

 ✓

C

a	b	1	1.2	e	34	2	2.5	6
---	---	---	-----	---	----	---	-----	---

 ✗

Declaration and Definition of 1D Array

Syntax: $\text{data_type name of the Array [no of elements]}$;

Example:- $\text{int Arr}[5];$

Arr

--	--	--	--	--

Compiler will allocate a Contiguous block of memory of Size = $5 \times \text{Size of (int)}$

* The length of an array can be Specified by any Positive Integer Constant expression.

$\text{int A}[5];$ ✓

$\text{int A}[5+5];$ ✓

$\text{int A}[5 \times 3];$ ✓

$\text{int a};$
 $\text{int A}[a = 10/3];$ ✓

$\text{int A}[-5];$ ✗

⇒ Specifying the length of an array Using macro is considered to be an excellent Practice.

```
#define N 10
```

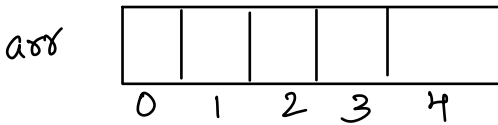
```
int A[N];
```

← This is macro

Accessing elements from 1D Array

To access an array elements, Just write

```
array_name[index];
```



Accessing the first element of an array: `arr[0]`

Accessing the Second element of an array: `arr[1]`

and So on....

⇒ Advantage of Using macros

* How to initialize one dimensional array?

Method 1:-

```
int A[5] = { 1, 5, -1, 0, 7 };
```

Method 2:-

```
int A[] = { 1, 5, -1, 0, 7 };
```

Method 3:-

```
int A[5];  
A[0] = 1;  
A[1] = 5;  
A[2] = -1;  
A[3] = 0;  
A[4] = 7;
```

Method 4:-

```
int A[5];  
for (i = 0; i < 5; i++)  
    scanf("%d", &A[i]);  
}
```

Q What if number of elements are lesser than the length specified?

```
int A[10] = {7, -1, 2, 4, 0, 7}
```

The remaining locations of the array are filled by Value 0.

```
int A[10] = {7, -1, 2, 4, 0, 7, 0, 0, 0, 0};
```

A simple tip

```
int A[10];  
for (i = 0; i < 10; i++) {  
    A[i] = 0;  
}
```



```
int A[10] = {0};
```



```
int A[10] = {}? ❌
```

Because, this is illegal.

⇒ you must have to specify at least 1 element if can not be completely empty.

⇒ and it is also illegal to add more elements than the length of an array.